



Indian Institute of Technology Gandhinagar

MA 104: ODE

Semester-II, Academic Year: 2025-26

Tutorial – 4

- (1) Consider the damped oscillator, driven by an external force $F_{\text{ext}}(t) = F_0 (1 - e^{-\beta t})$, so that the ODE is

$$my'' + cy' + ky = F_0 (1 - e^{-\beta t}),$$

where $\beta > 0$. Find the solution for the underdamped case assuming $y(0) = 0$ and $y'(0) = 0$. Analyze your solution for large β (external force quickly jumps to F_0) and very small β (external force turns on slowly).

- (2) (*Higher order Cauchy-Euler equations*) Let $D = \frac{d}{dx}$ and $D = \frac{d}{dt}$ where $x = e^t$. Show

$$xD = D, \quad x^2 D^2 = D(D-1), \quad x^3 D^3 = D(D-1)(D-2).$$

Hence conclude that $(x^3 D^3 + ax^2 D^2 + bxD + c)y = 0$ for $x > 0$ is transformed into the constant coefficient ODE

$$[D(D-1)(D-2) + aD(D-1) + bD + c]y = 0$$

by the substitution $x = e^t$.

- (3) (*Higher order variation of parameters*) Consider the n -th order linear equation

$$y^{(n)} + \sum_{i=0}^{n-1} a_i(x)y^{(i)} = r(x).$$

Assume that $y_1, \dots, y_n$ are linearly independent solutions of the associated homogeneous equation. Prove that a particular integral is

$$y_p = \sum_{i=1}^n v_i y_i, \quad \text{where} \quad v'_i = \frac{R_i}{W}.$$

Here W is the Wronskian of $y_1, \dots, y_n$ and R_i is the determinant obtained by replacing the i -th column of W by $[0, 0, \dots, 0, r(x)]^T$.

- (4) Solve the following non-homogeneous linear ODEs using variation of parameters and the method of undetermined coefficients.

(a) $y'' - 4y' + 5y = e^{2x} \csc x$

(b) $y'' + 6y' + 9y = \frac{16e^{-3x}}{x^2 + 1}$

- (c) $y'' - y = \frac{1}{\cosh x}$
 (d) $y''' - y'' + y' - y = e^{-t} \sin t$
 (e) $y''' + y' = \sec t, \quad -\frac{\pi}{2} < t < \frac{\pi}{2}.$

(5) Solve the IVP:

$$x^3 y''' + xy' - y = x^2, \quad \text{with} \quad y(1) = 1, \quad y'(1) = 3, \quad y''(1) = 14.$$

(6) Determine intervals in which solutions are sure to exist.

- (a) $y^{(4)} + 4y''' + 3y = t^2$
 (b) $t y''' + (\sin t) y'' + 3y = \cos t$

(7) Use the method of reduction of order (i.e., $y = u(x)y_1(x)$) to solve the differential equation

$$(2 - t)y''' + (2t - 3)y'' - ty' + y = 0, \quad t < 2; \quad y_1(t) = e^t.$$

(8) Determine whether the given functions are linearly dependent or linearly independent.

- (a) $f_1(t) = 2t - 3, \quad f_2(t) = 2t^2 + 1, \quad f_3(t) = 3t^2 + t.$
 (b) $f_1(t) = 2t - 3, \quad f_2(t) = t^2 + 1, \quad f_3(t) = 2t^2 - t, \quad f_4(t) = t^2 + t + 1.$

(9) Show that $W(5, \sin 2t, \cos 2t) = 0$ for all t . Can you establish this result without direct evaluation of the Wronskian?

(10) Let $y_1(x)$ and $y_2(x)$ be two linearly independent solutions of

$$x^2 y'' + 5xy' + 4y = x^2 \sin x.$$

Given $W(y_1(2), y_2(2)) = \frac{1}{32}$, find the Wronskian $W(y_1(x), y_2(x))$.

(11) Consider the ODE

$$y'' + \frac{1 - \alpha^2}{4x^2} y = 0.$$

Comment on the nature of the solution for $\alpha = \frac{1}{n}$.

(12) Find the general solution of

- (a) $y'' - y' - 2y = 4x^2$
 (b) $y'' + 10y' + 25y = 14e^{-5x}$

(13) (a) Show that applying variation of parameters to $y'' + y = f(x)$ leads to the solution

$$y_p(x) = \int_0^x f(t) \sin(x - t) dt.$$

(b) Find a similar formula for a particular solution of

$$y'' + k^2y = f(x), \quad \text{where } k > 0.$$

(14) Find the general solution of:

(a) $(1-x)y'' + xy' - y = (1-x)^2$

(b) $xy'' - (1+x)y' + y = x^2e^{2x}$

(c) $y'' + 4y = 2\cos 2x + 10e^x$

(d) $y'' - y = e^{-x}(\sin x + \cos x)$

(e) $y''' - 3y'' - y' + 3y = x^2e^x$

(15) Let $y_1(x), y_2(x)$ be linearly independent solutions of

$$y'' + p(x)y' + q(x)y = 0.$$

Show that $\phi(x) = \alpha y_1(x) + \beta y_2(x)$ and $\psi(x) = \gamma y_1(x) + \delta y_2(x)$ are linearly independent if and only if $\alpha\delta \neq \beta\gamma$.

(16) ¹Show that any nontrivial solution $u(x)$ of

$$u'' + q(x)u = 0, \quad q(x) < 0 \quad \forall x,$$

has at most one zero.

(17) Let $u(x)$ be a nontrivial solution of

$$u'' + [1 + q(x)]u = 0, \quad q(x) > 0.$$

Show that $u(x)$ has infinitely many zeros.

(18) Let $u(x)$ be a nontrivial solution of

$$u'' + q(x)u = 0$$

on a closed interval $[a, b]$. Show that $u(x)$ has at most finitely many zeros in $[a, b]$.

¹For (16) & (17), use the *Sturm Comparison Theorem* [p. 194, Simmons].